Day 2 — CNNs

Building, training, transfer-learning, and explaining

Deep Learning for Computer Vision — theory

Before we build, read **cs231n Module 2**:

https://cs231n.github.io/

It covers convolutional layers, pooling, spatial arithmetic, and the classic architectures (AlexNet, VGG, ResNet). ~1–2 hours.

We'll work from these concepts all day — having them fresh makes the hands-on parts click faster.

Today's notebooks

- `04_first_cnn.ipynb` — build a CNN from scratch
- `05_training_cnn.ipynb` — full training loop, monitor loss & accuracy
- `06_transfer_learning.ipynb` — fine-tune a pretrained model
- `07_gradcam.ipynb` — see *what* the model is looking at

By the end of the day you'll have trained your own CNN and visualized its attention on real images.

Let's build.

Open `notebooks/04_first_cnn.ipynb`